

Python Serial Communication, AN007

In this application note we create a Python user defined function named NT_Serial, shown to the right, in order to send commands to a Novatech Instruments Model 409B Signal Generator and display the response returned by the 409B. We are using the Pycharm IDE and Pyserial library. An explanation of the code is as follows:

After comment **#A** we define a function named NT_Serial with parameters “com” and “command”.

After comment **#B** we import the serial class from the Pyserial library, instantiate a serial class object named “tep” and then load the “tep” object with serial port parameters.

After comment **#C** we open the “tep” serial port. If there is a problem opening the port we print an error message.

After comment **#D** we write the contents of the “command” variable to the “tep” serial port. The “encode()” method creates HEX values for these characters using the UTF-8 code set. Then the tep.readline() method stores the first line returned from the COM port and stores it in the “temp” variable. The “decode()” method converts the returned HEX values to text characters, also based on the UTF-8 code set.

After comment **#E** we create a variable “c” and a variable “recentVal”. The tep.readline() method stores the second line returned from the COM port (if there was more than one line returned) and stores this in “recentVal”. The while loop checks to see if there was more than one line returned. If more than one line was returned then the lines are stored in “temp” separated by a “c” string. The temp.split(c) method removes the “c” strings, separates the lines with commas and prints them to the screen.

After comment **#F** we load the NT_Serial() function with the correct COM port and 409B command.

If we run Pymain in the IDE and if we have a COM3 port on our computer that is connected to a Model 409B, then the 409B will generate a 15MHz sine wave output. If the 409B echo is on, the following string is returned to the computer screen:

"f0 15/n", "OK"

Pymain code

```
#A
def NT_Serial(com, command)
#B
    import serial
    tep = serial.Serial()
    tep.port = com
    tep.baudrate = 19200
    tep.parity = serial.PARITY_NONE
    tep.stopbits = serial.STOPBITS_ONE
    tep.timeout = 1
#C
    try:
        tep.open()
    except Exception as e:
        print("error opening serial port: \n" + str(e))
#D
    tep.write(command.encode())
    temp = tep.readline()
    temp = temp.decode()
#E
    c = "!"$"
    recentVal = tep.readline()
    recentVal = recentVal.decode()
    while (recentVal != ""):
        temp = temp + c + recentVal
        recentVal = tep.readline()
        recentVal = recentVal.decode()
    Lines = temp.split(c)
    m = lines
    print(m)
#F
NT_Serial("COM3", "f0 15\n")
```